Assignments

Combinatorics

Fall 2026

Written homework

Assignment	Due	Solutions
Homework 1	Thursday, August 27, 2026 23:58	Solutions
Homework 2	Thursday, September 3, 2026 23:58	Solutions
Homework 3	Thursday, September 17, 2026 23:58	Solutions
Homework 4	Thursday, September 24, 2026 23:59	Solutions
Homework 5	Thursday, October 1, 2026 23:59	Solutions
Homework 6	Thursday, October 8, 2026 23:59	—
Homework 7	Thursday, October 29, 2026 23:59	—
Homework 8	Thursday, November 5, 2026 23:59	—

Submit your work through Gradescope using Blackboard's assignment area. The printable version is available from each assignment page.

Weekly homework gives you a chance to work through more involved problems using the techniques we have learned. A complete solution includes both a clear explanation and correct computations. Every exercise is required; point values indicate how much each exercise contributes to the assignment grade.

See Example Homework Write-Ups for two sample exercises with sample rubric, full-credit, and partial-credit solutions.

- Please write your homework by hand.
- For each exercise, start a new page.
- Scan your homework to PDF for submission.
- Submit it on Gradescope and clearly mark the pages associated with every exercise.

Solutions will be posted here within one week of the submission deadline. The deadlines will also appear in the Blackboard Calendar.

Practice worksheets

Optional practice worksheets help you prepare for an exam. They are not collected or graded, and each one has posted solutions.

Midterm 3. See also the Midterm 3 information page.

- Worksheet 1: Groups and Permutations (solutions)

- Worksheet 2: Orbits, Stabilizers, and Fixed Sets (solutions)
- Worksheet 3: Rotations of Solids (solutions)
- Worksheet 4: Burnside's Lemma (solutions)

Online questions

Online questions usually follow each lecture and are due at the start of our next class meeting. They are available through Blackboard's online-question area; deadlines appear in the Blackboard Calendar.

Online questions provide a quick chance to practice vocabulary and core skills before we build on the material. Most questions allow multiple attempts.

Academic integrity, collaboration, and artificial intelligence

You are encouraged to work with classmates, tutors, and me: talking through ideas, definitions, and problem-solving strategies is a valuable part of learning. You must understand every solution you submit and write it based on your own understanding. Please do not copy another person's solution or write-up.

You may use generative artificial intelligence as a study aid, but not to produce any work that you submit for credit. It is fine to ask a study question such as, "I do not understand when to use generating functions to count combinations with restrictions." It is not acceptable to ask an AI system how to solve a specific assigned problem. Rephrasing a generated solution using your own words is not acceptable. If you are unsure whether a particular use is allowed, ask me before you submit your work.

When you submit an assignment, disclose any collaboration or use of external tools, even when it is permitted.